Technology Usage Facet

The Short TL;DR Version

Thinking of using a technology that is **new to your project**? Talk to Architecture first – wherever it sits on this list.

Already running something in **Prohibited** or **Immediate Action Required**? Talk to us – it needs a plan to move off.

An **Approved** option not quite fitting your need? There may be a better fit – come talk to us.

The Detail

The **Technology Usage Facet (TUF)** is part of the Solera Application Scorecard, developed and maintained by the Global Architecture group. It guides, scores and enforces technology-usage standards across Solera applications, and replaces the former Tech Radar. It is reviewed and adjusted regularly together with stakeholders.

TUF classifies the technologies we use for **new and continuing development**. It describes **fit for use, not fit for purpose** – a tool can be approved for general use yet still be the wrong choice for a specific job. Check with Architecture if you are unsure a tool fits your purpose.

Below is the classification used for technologies across both new and continuing development initiatives:

Emerging – Under trial or Proof of Concept. May not be production-ready or well supported. Architectural review **and** approval are required before use on a project.
Approved – Corporate standard; strongly encouraged for new development and upgrade / migration work. Architectural review is required; approval is granted implicitly.
Restricted – Well-supported and actively used at Solera, with good in-house knowledge. Maintenance / feature work needs review if it affects scale, usability or how the technology is used; **new** use needs review **and** approval.
Prohibited – Treat as technical debt and actively migrate away. Applications using these are considered out of compliance.
Immediate Action Required – At risk. Remove before adding any new features; architectural approval will **not** be granted.

IMPORTANT: If you do not see a tool / software you want to use in the tables below, reach out to DL-WW-GIT-Architecture to discuss your options **BEFORE** you commit to using it.

CLOUD AND NON-CLOUD ON-PREM EQUIVALENCES: Cloud DevOps Technology Usage Facet and Cloud Native and not Cloud Technologies to Solera On-premise Equivalence

Architectural Review / Approval Matrix

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Greenfield development	• Do not use • Architectural review required • Architectural approval required	• Usage encouraged • Architectural review required • Architectural approval not required	• Usage discouraged • Architectural review required • Architectural approval required	• •	• •
New feature development	• Do not use • Architectural review required • Architectural approval required	• Usage encouraged • Architectural review required • Architectural approval not required	• Usage discouraged • Architectural review required • Architectural approval required	• • •	• •

Technology Categorization Matrix

Technologies are grouped into domains below. Within each domain table, the five rating columns keep their standard colours.

AI & Machine Learning

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required

Generative AI & LLMs	—	Azure OpenAI M365 Copilot Solera LLM Platform "Ashley" (product integration) AI Foundry (oai.azure.com, ad-hoc use) GitHub Co-Pilot	—		
AI Coding & Agentic Tools	MCP (Model Context Protocol)	GitHub Copilot GitHub (web)	—		
AI Media & Image Generation	—	Remove.bg DALL·E 3	—		
Document Intelligence / OCR	—	Azure AI Document Intelligence	LEADTOOLS Accusoft		

Data & Analytics

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Relational Databases		MS SQL Server 2025	Oracle PostgreSQL MS SQL Server 2022		
NoSQL & Object Databases	Apache Cassandra	MongoDB CosmosDB (Cloud) DynamoDB (Cloud)	—		
Caching	—	Redis	—		
Search	—	OpenSearch ElasticSearch 7.17.x	Solr Sphinx		
Data Management & Governance	Apache Nutch Azure Purview Great Expectations OneTrust (Data Discovery)	Airflow (data pipeline)	Hadoop MQSeries Presto		
ETL & Data Processing	—	MS SQL Server Integration Services (SSIS)	Spark StreamSets Pentaho Data Integration (Kettle)		

Analytics & Reporting	—	MS SQL Server Analysis Services (SSAS) MS SQL Server Reporting Services (SSRS)	Power BI (incl. embedded) Tableau (incl. embedded) *		

Integration & Messaging

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
API Gateway & Management		Solera API HUB OpenAPI (Swagger) Kong API Gateway (Tanzu use) Azure API Management Cloud-native API gateways (AWS, GCP, Azure)	TIBCO Mashery		
Communication Protocols	GraphQL	REST HTTPS SignalR gRPC	ASP.NET XML Web Services / SOAP WebSocket		
Messaging & Event Streaming	—	RabbitMQ AWS MSK	Apache Kafka * NServiceBus		
Notifications (Email / SMS / Push)	Azure Notification Hubs (push)	Microsoft Purview – PII-email feature (specific secure delivery method; transactional, non-marketing, GLBA-compliant) SendGrid (**marketing emails only**) Azure Communication Services (SMS)	Twilio (SMS)		

DevOps & SDLC

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Source Control	—	GitHub Enterprise	—		
CI/CD – Build	SCM-managed JFrog	Jenkins Azure Pipelines GitHub Actions	—		
CI/CD – Deployment	—	GitHub Actions Jenkins Azure Pipelines	Octopus (some regional licensing exceptions)		

Artifact Repositories	—	AWS ECR (Cloud) Harbor Nexus SCM-managed JFrog	Klondike ProGet (being retired JFrog)		
Code Quality & Security Scanning	—	SonarQube Contrast	—		
Test Management	—	TestRail Pro (On-prem or Cloud)	—		
Infrastructure as Code & Config Automation	—	Terraform / OpenTofu Ansible	Azure Resource Manager		

Infrastructure & Operations

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Compute, Hosting & Virtualisation	—	Tanzu (consult architecture) Rancher – platform (consult architecture) VMware (on-prem)	AWS Azure Hyper-V (third-party DC virtualisation)		
Containers & Orchestration	—	Tanzu Rancher Desktop	Docker Desktop (licence required) K3s (third-party vendor req) Minikube (third-party vendor req)		
Web & Application Servers	—	IIS Tomcat Kestrel (.NET in Linux containers) Apache HTTP Server (static content)	JBoss WildFly NGINX (static content)		
Load Balancing, Reverse Proxy & Web Proxies	—	F5 (physical or virtual) NGINX (Kubernetes) Cloud-native load balancer CloudFront	HA Proxy NGINX (VM) Apache (load balancing) Bright Data proxy services		
Job Scheduling	—	Windows Task Scheduler cron (Linux) Kubernetes CronJob Rundeck	ActiveBatch Hangfire VisualCron		
Observability (Logging, Monitoring & Alerting)	Zabbix	Graylog New Relic Icinga Observium Serilog Graylog (C#) CloudWatch (Cloud) PagerDuty	log4net Nlog Prometheus Grafana DataDog		

Disaster Recovery	—	*Solera DCs* Zerto (for VMware vSphere) Dell PowerMax SRDF Dell Unity Replication Technologies Dell ECS Replication Groups Dell Data Domain Replication	—		
File Storage	—	AWS S3 (Cloud) Azure Storage Services (Cloud) Dell Unity NAS (NFS/SMB) Dell ECS Object / S3-compatible Dell Data Domain Backup	MinIO SQL Server FILESTREAM SQL Server VARBINARY SMB v1		
GIS, Geolocation & Routing	Azure Maps	Google Maps API (caution if app used in China)	ESRI ArcGIS Trimble Maps / PC*MILER		

Identity & Security

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Authentication, Authorization & Identity	—	**SOLID / GP2 (Azure B2C)** Entra ID (internal apps / client credentials)	Ping Identity Keycloak AWS Cognito		
Secrets Management	—	AWS Secrets Manager (Cloud) Azure Key Vault (Cloud) GCP Secret Manager (Cloud) HashiCorp Vault (On-Prem)	—		

Languages, Frameworks & Front-End

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required

Languages & Frameworks	GoLang R Scala Testcontainers	.NET (Core) latest LTS (currently 10) C# Java – Temurin latest LTS, Amazon Corretto >=25 for AWS. TypeScript React (with TypeScript) Bootstrap React testing: Jest, React Testing Library, Cypress	.NET Framework 4.6.2 Angular 17 Java non-latest LTS (OpenJDK) Python latest (3.12) JavaScript (prefer TypeScript) ASP.NET Web Forms ASP.NET MVC ASP.NET Core Razor Pages Node.js Blazor		
Mobile Development	Kotlin Multiplatform	Swift (Apple)	—		
UI/UX & Design Systems	—	Storybook (when engaging the UI/UX team) Solera global-design-storybook Ant Design / React-Bootstrap (optional, if not using the UI/UX team)	Building your own UI/UX framework		
Object Mapping	—	Manual mapping	—		

Tooling, Platforms & Business Software

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
.NET Tools & Libraries	—	Xceed Zip for . NET Reporting Services (PDF generation) NUnit Dapper (needs architecture review)	iTextSharp PdfSharp Aspose.Cells ABCpdf LEADTOOLS NetGender BCC Software / Satori jQuery WebGrease Yahoo UI Compressor EcmaScript.NET Newtonsoft.Json Owin Microsoft.Owin Polly.Caching. IDistributedCache CSLA		
Java Tools & Libraries	—	IntelliJ IDEA JUnit Gradle 9+ Spring Boot	Gradle < 8 Maven 3.6.3+		

JavaScript Tools & Libraries	—	CKEditor 5 (commercial licence) TinyMCE 7 (commercial licence)	CKEditor 4 TinyMCE 6 jQWidgets		
SQL & Database Tooling	—	SSMS Azure Data Studio	RedGate SQL Prompt Professional		
Developer Utilities	Split (feature flags)	Draw.io Jupyter Notebook Fiddler Classic Postman PuTTY Notepad++ Visual Studio (Pro/Enterprise) VS Code JetBrains Tools Visio Cloud (O365) Selenium PlayWright	Visio Desktop (only with an existing assigned licence) Microsoft Power Automate (workflow / RPA)		
Visual Assets & Icons	—	FontAwesome	—		
Operating Systems – Linux	RHEL 10 (Nov 2026)	RHEL 9 Amazon Linux 2023	RHEL 8 Ubuntu 22.04 (team owns responsibility, supportability, patching)		
Operating Systems – Windows	—	Windows Server 2025 Windows 11	Windows Server 2022		
File Sharing	—	NA – contact Architecture EMEA – GFSS Standard (JSCAPE)	WS_FTP (legacy, SFTP /FTPS only) ProFTPD FTPD Cerberus FTP SolarWinds Serv-U Microsoft OneDrive (no HA) WinSCP FileZilla Native OS SFTP/FTP		
Translation Tools	—	Unbabel (customer support) Trados Studio (Autodata)	—		

E-Signature	—	Adobe Sign (embedded) DocuSign (standalone)	—		
Product Analytics & Feedback	—	—	Pendo.io Google Analytics MixPanel Amplitude		

Encryption & Transport Security

	Emerging	Approved	Restricted	Prohibited	Immediate Action Required
Transport Layer Security	—	TLS 1.2 TLS 1.3	—		
Cipher Suites – TLS 1.2	—	ECDHE-ECDSA / RSA-AES128 / 256-GCM CHACHA20-POLY1305 variants	DHE-RSA-AES128 / 256-GCM		
Cipher Suites – TLS 1.3	—	TLS_AES_128_GCM_SHA256 TLS_AES_256_GCM_SHA384 TLS_CHACHA20_POLY1305_SHA256	CCM suites (embedded only)		
Data Transfer over Public Internet	QUIC	HTTPS SFTP IPsec VPN IP allow-listing (encrypted)	FTPS		

Footnotes

***** Under review.
****** Allowed with an architecture waiver (not considered technical debt).